

Customer Shopping Behavior Analysis

Transactional insights from **3,900 purchases** across product categories — uncovering spending patterns, segments, and strategic opportunities.



Dataset at a Glance

3,900

Transactions

Total purchase records

18

Columns

Demographics, purchases, behavior

37

Missing Values

In Review Rating only

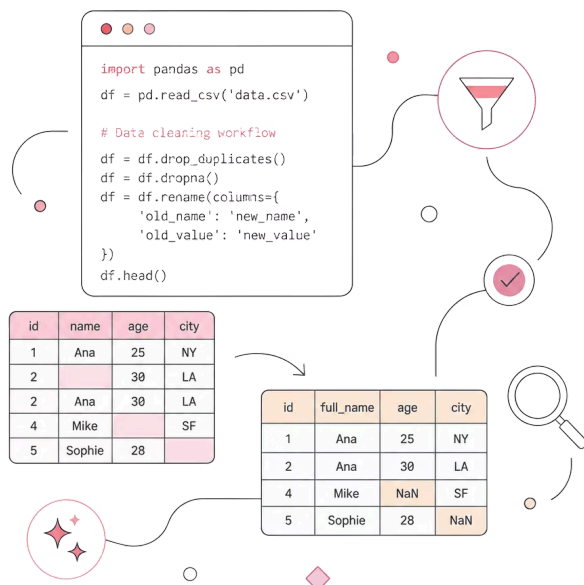
50

Locations

Geographic coverage

Key features: Age, Gender, Subscription Status, Purchase Amount, Category, Discount, Shipping Type, Review Rating

Data Preparation in Python



→ Load & Explore

Imported via pandas; `df.info()` & `.describe()`

→ Clean & Standardize

Imputed Review Rating by category median; renamed to `snake_case`

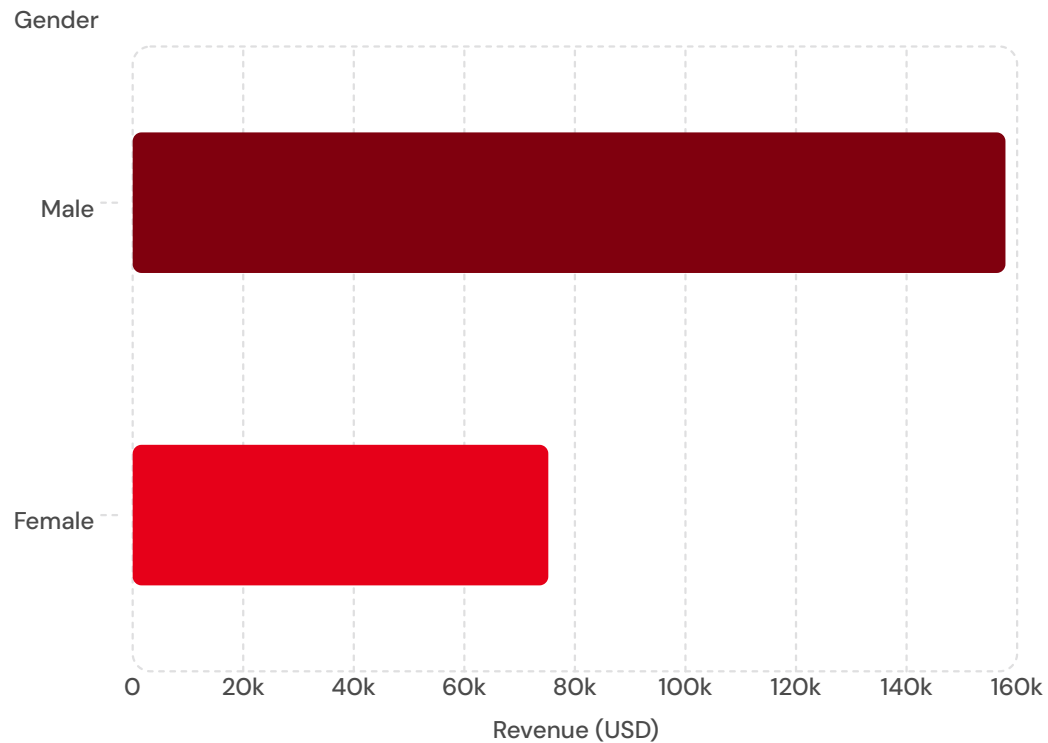
→ Engineer Features

Created `age_group` & `purchase_frequency_days`; dropped redundant `promo_code_used`

→ Load to PostgreSQL

Cleaned DataFrame loaded for SQL analysis

Revenue by Gender



Key Insight

Male customers generated **2.1× more revenue** than female customers (\$157,890 vs \$75,191).

- ❑ Male purchases: 2,652 vs Female: 1,248 — volume drives the gap.



Discounts & High Spenders

839 Customers

Used discounts AND spent above average purchase amount

Top Discounted Products

Hat (50%), Sneakers (49.7%), Coat (49.1%) — highest discount rates

Shipping Premium

Express avg \$60.48 vs Standard \$58.46 — small but consistent gap

Subscribers vs Non-Subscribers

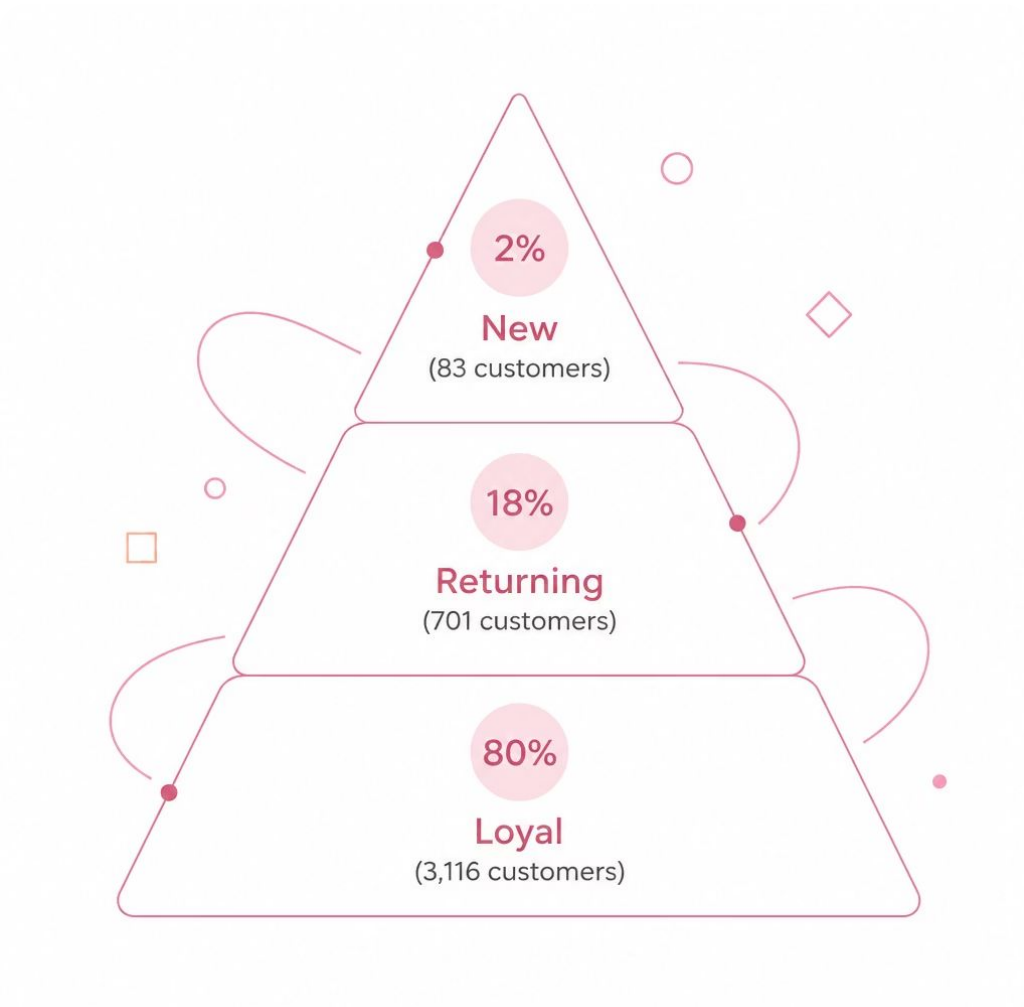


The Subscription Gap

Only **27% subscribed** (1,053 customers), yet avg spend is nearly identical: \$59.49 vs \$59.87.

- ❑ 958 repeat buyers (>5 purchases) are NOT subscribed — major conversion opportunity.

Customer Segmentation



Segment Breakdown

Customers classified by purchase history into three tiers.

Loyal — 3,116

80% of base

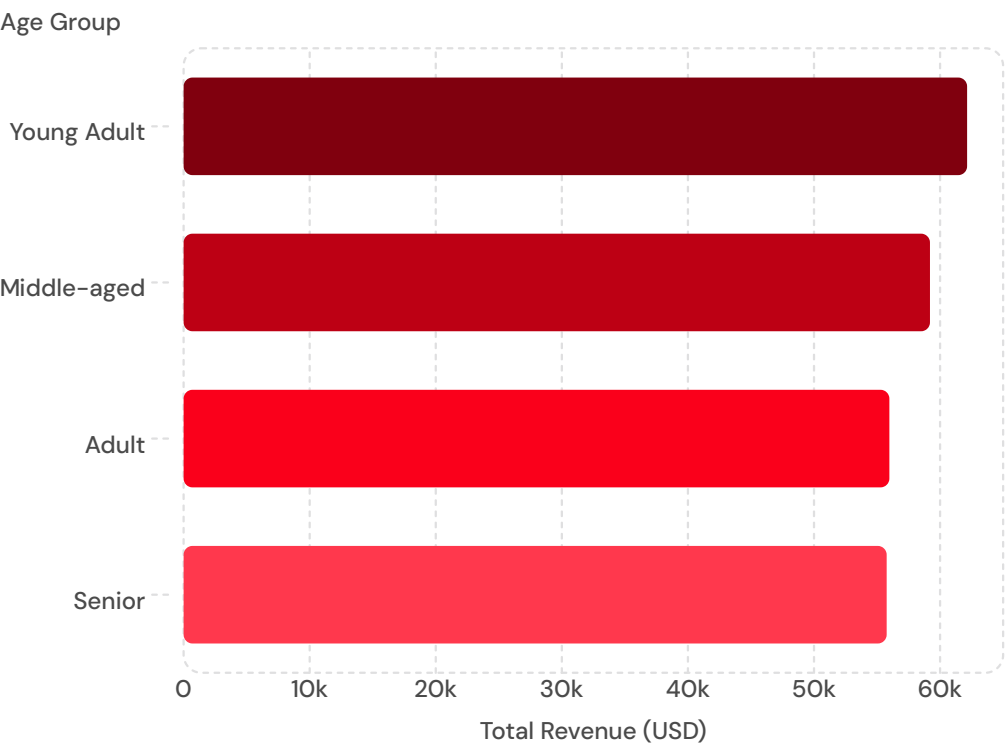
Returning — 701

18% of base

New — 83

2% of base

Revenue by Age Group & Top Products



Top-Rated Products

- 01

Gloves

3.86★
- 02

Sandals

3.84★
- 03

Boots

3.82★
- 04

Hat

3.80★
- 05

Skirt

3.78★

Power BI Dashboard



Dashboard Highlights

3.9K Customers

Avg purchase \$59.76

**Clothing
Dominates**

\$100K revenue, 1,800
sales

Filters

Subscription,
Gender, Category,
Shipping

Business Recommendations



Boost Subscriptions

Promote exclusive benefits; target 958 unsubscribed repeat buyers



Loyalty Programs

Reward Returning customers to move them into Loyal segment



Review Discount Policy

Balance sales lift with margin control on high-discount items



Targeted Marketing

Focus on Young Adults, express-shipping users, and top-rated products

